

Predictive Model Plan

Task 2 — Predicting Delinquency with AI

1. Model Logic (Generated with GenAI)

Objective: predict the probability that a customer will be classified as delinquent (Delinquent_Account = 1) so Geldidium's Collections team can prioritize appropriate early interventions. The supplied dataset contains 500 customer records and 19 variables, with 80 delinquent accounts (16.0%). The recommended conceptual approach is Logistic Regression as the primary model, with a Decision Tree as a challenger model.

Top 5 input features

- Credit_Utilization — candidate indicator of credit pressure; EDA showed slightly higher average utilization among delinquent customers.
- Debt_to_Income_Ratio — candidate indicator of repayment burden; EDA showed a slightly higher average among delinquent customers.
- Credit_Score — candidate measure of creditworthiness; predictive strength should be validated through modeling.
- Income — candidate measure of repayment capacity; missing values require documented treatment.
- Loan_Balance — candidate measure of outstanding debt exposure; missing values require documented treatment.

Conceptual workflow

- Define Delinquent_Account as the binary prediction target.
- Standardize categorical values, handle missing numerical values using training-set-only imputation, and validate anomalous Credit_Utilization values.
- Select the five primary candidate features and test additional validated features only if they improve out-of-sample performance without leakage.
- Use a stratified train/validation/test design so the minority delinquency class is represented consistently.
- Train Logistic Regression as the primary baseline and Decision Tree as a challenger.
- Generate a delinquency probability, then select an operating threshold based on the Collections team's business costs and intervention capacity.
- Assign risk priority and route customers to appropriate human-reviewed interventions.

Simplified model logic

- Customer inputs → preprocessing → trained Logistic Regression → probability of delinquency → validated threshold → risk priority → human-reviewed intervention.
- Do not use the score as an automatic adverse action; it should support prioritization and remain subject to governance and review.

2. Justification for Model Choice

Logistic Regression is recommended because it produces an interpretable probability of delinquency and its coefficients can be reviewed to explain how features contribute to risk. This is well suited to financial services, where transparency, auditability, monitoring, and consistent decision processes are important. A Decision Tree is a useful challenger because it can capture nonlinear relationships and interactions while remaining relatively interpretable, but it may overfit without careful tuning. The final model should be selected using out-of-sample performance, calibration, stability, and fairness—not accuracy alone.

Why the choice fits Geldium

- Clear probability output supports collections prioritization.
- Coefficients provide a transparent basis for explaining model behavior.
- Relatively simple to implement, monitor, recalibrate, and govern.
- Provides a strong interpretable baseline for comparison with more complex models.
- Supports responsible AI requirements such as explainability and human oversight.

3. Evaluation Strategy

Because the dataset has a 16% delinquency rate, evaluation should not rely on accuracy alone. Use stratified train/validation/test splits and evaluate ranking performance, classification quality, probability calibration, and fairness before operational use.

Key evaluation metrics

- ROC-AUC — measures how well the model ranks delinquent customers above non-delinquent customers.
- Precision — measures how many customers flagged high-risk are actually delinquent; useful for controlling unnecessary outreach.
- Recall — measures how many delinquent customers are identified; important for reducing missed intervention opportunities.
- F1 Score — summarizes the precision–recall trade-off.
- Confusion Matrix — shows false positives and false negatives at the selected operating threshold.
- Calibration — checks whether predicted probabilities correspond to observed delinquency rates.

Fairness, bias, and ethical safeguards

- Compare recall, precision, false-positive/false-negative rates, and calibration across relevant customer groups where legally and operationally appropriate.
- Investigate material disparities and assess whether they arise from data quality, feature selection, sampling, or model behavior.
- Protect customer confidentiality and avoid exposing unnecessary personally identifiable or sensitive financial information to GenAI tools.
- Maintain human oversight for collections decisions and document model limitations, intended use, and monitoring procedures.
- Use the model to prioritize supportive, proportionate interventions rather than automatically penalizing customers.

GenAI Support

GenAI was used to scaffold the conceptual modeling approach, compare model options, identify relevant evaluation metrics, and structure the fairness and governance plan. The outputs should be validated against the supplied dataset and business requirements before implementation.

Prompts used

- Outline a predictive modeling pipeline to forecast credit delinquency, from feature selection to model evaluation.
- Suggest 2 modeling options (simple and complex) for predicting delinquency, and recommend one.
- Explain how a delinquency risk model transforms customer input variables into a final risk prediction, from data ingestion to prediction output.
- Suggest a set of metrics to evaluate a financial risk prediction model for fairness, bias, and accuracy.

Final recommendation

Use Logistic Regression as the primary conceptual model and Decision Tree as a challenger. Proceed to implementation only after the Task 1 data-quality issues are addressed, the prediction window is defined to prevent target leakage, and the selected model demonstrates adequate predictive performance, calibration, stability, and fairness.



THANK YOU